

Potions Live Platform - External Scrutiny Reference

Version: 0.3

Date: 2026-06-30

Audience: External quants, developers, allocator diligence reviewers

Scope: Platform machinery only - not strategy rule definitions. Promotion status, rule-family names, and proprietary signal definitions are intentionally kept in internal research trackers. For validation design see [live/specs/CAUSAL_VALIDATION_MASTER_SPEC.md](#). For audit pass/fail see [data/docs/AUDIT_TRACKER.md](#).

1. Executive map

```
mermaid
flowchart LR
  Raw[Raw 1m DBN/CSV or daily CSV] --> Bars[Bars[bars.py dense RTH]]
  Bars --> Engine[Engine[Engine.process_bar]]
  Engine --> Plugin[Plugin[StrategyPlugin]]
  Plugin --> Intent[Intent[OrderIntent]]
  Intent --> Broker[Broker[PaperBroker]]
  Broker --> Store[Store[FlatFileStore]]
  Store --> Audit[Audit[replay audit / drivers]]
  Audit --> Metrics[Metrics[institutional_strategy_metrics]]
  Metrics --> Score[Score[generate_strategy_validation_scorecard]]
```

Paper-first v0: Strategies default to paper mode. TradovateBroker / CQGBroker are boundary shells; live routing is not production-ready. State persists under [live/state/](#) as flat CSV/JSON.

2. Assumptions and non-goals

Assumption	Implication
OHLC bars hide intra-bar path	Multiple valid fill outcomes per bar; broker picks one deterministically (stop-first, gap-through)
Completed bar before signal	Daily plugins evaluate only closed bars; intraday uses 1m bar close
live_after_ts on orders	Fills cannot occur on or before activation timestamp
Conservative replay defaults	1-tick slippage, fees, spread model in hardened replays (replay_realism.py)

Not guaranteed: Sub-minute execution ordering from 1m bars alone; live/paper parity; peer-benchmark DSR without sourced factsheets.

3. Candle and bar construction

Module: live/bars.py

Dense RTH 1m grid (09:30-15:59 America/New_York). Missing minutes forward-fill open=high=low=close=prev_close, volume=0:

```
python
def dense_rth_1m_bars(df, session_day):
    grid = rth_minute_index(session_day) # 390 minutes
    # ... reindex sparse Databento rows onto grid, forward-fill ...
```

Causality: Indicators and plugins must only use bars with ts <= current_bar.ts. Replay drivers load history up to session date; no future-day bars in the bar loop.

Timezone: Session logic uses America/New_York. Stored timestamps are ISO with offset.

4. Engine orchestration

Module: live/engine.py

Per-bar loop:

1. Optionally persist bar to store.
2. PaperBroker.process_bar(bar) - may produce fills (up to 20 inner iterations for cascading fills).
3. StrategyManager.on_fills(fills) - strategy reactions.
4. StrategyManager.on_bar_close(bar) - signal generation, new intents.
5. Repeat broker pass for intents submitted on bar close.
6. Optional process_market_close_bar for scheduled MOC flattens.

```
python
def process_bar(self, bar: Bar) -> None:
    for _ in range(20):
        fills = self.broker.process_bar(bar)
        if not fills:
            break
        self.manager.on_fills(fills)
    self.manager.on_bar_close(bar)
```

5. Strategy plugin contract

Modules: live/registry.py, live/models.py, live/strategies/*.py

Plugins implement bar-close callbacks and emit OrderIntent objects (side, type, qty, prices, live_after_ts, brackets). The engine does not embed strategy rules - only routing, persistence, risk, verification.

Tier-1 plugin classes (validation focus):

Plugin class	Bar frequency	Validation focus
Intraday opening-range engine	Intra-day	Order timing, OCO/bracket lifecycle, delayed arming
Higher-timeframe breakout engine	Daily	Daily bar causality, resting order lifecycle
Trend-following accumulation engine	Daily / weekly per config	Multi-bar feature availability, stack state, exits

6. Broker and fill semantics

Module: live/broker.py

live_after_ts

Non-market orders do not fill until bar.ts is strictly after order.live_after_ts:

```
python
if order.live_after_ts and not _ts_after(bar.ts, order.live_after_ts):
    continue # skip fill this bar
```

Stop-first OCO ambiguity

Active orders sorted so stops evaluate before limits within the same bar - pessimistic for protective exits:

```
python
return sorted(snapshot, key=lambda oid: (
    _FILL_PRIORITY.get(self._orders_cache[oid].order_type, 9), ...
))
```

Gap-through stops

Stop fills can use bar open when price gaps through the stop (validated in broker_realism_validation.py).

Slippage and spread

Market/stop fills apply slippage_ticks adverse; optional SpreadModel widens fills in RTH open / low volume.

Validation: live/broker_realism_validation.py - 10+ known-answer cases with chart artifacts under live/state/broker_realism_validation/.

7. Causal ordering

Feature-level point-in-time audit

- live_after_ts - activation time on every intent/order.
- Plugin gates - prerequisite event maps must list required same-session events before arming.
- FeatureSnapshot rows - opt-in Tier-1 plugins persist the feature facts used at decision time with event_ts, available_at_ts, and current_bar_ts.
- CausalityGuard - validates event ts <= available_at_ts <= current_bar_ts; in strict mode it blocks new non-reduce orders on violation while allowing protective reduce-only exits.

Current Tier-1 emitters: selected intraday, hourly, higher-timeframe breakout, and trend-following plugins. Regenerated replay states now include feature_snapshots.csv and header-only causality_violations.csv when no violations occurred.

Campaign-level gate audit

Module: internal delayed-arming replay driver - validate_prior_condition_entries():

After replay, for each entry fill, checks that the required same-calendar-day prerequisite event existed with event_ts < entry_ts. Increments causality_violations if missing. This sits beside the FeatureSnapshot audit: the campaign audit proves the delayed gate existed before the campaign fill; the feature audit proves each recorded decision feature was available when consumed.

Current regeneration status

2026-06-29 regeneration produced 57 feature_snapshots.csv files under live/state/ and 0 actual causality violation rows. Coverage includes:

- Delayed-arming intraday replay states.
- Ungated intraday OCO replay states.
- Hourly prerequisite-event replay states.
- All 42 generated broker-like daily replay states, including higher-timeframe breakout and trend-following rows.

One market-specific delayed-arming row remains banked from an earlier strict replay but was not regenerated in this pass because the corresponding local 1m source is missing.

Execution ambiguity (1m limits)

Module: internal execution-scrutiny driver

Classifies ambiguous_same_1m_bar and pre_arm_breakout_touch. Routes rows to tick_replay_manifest.csv per market. Tick reconstruction not complete at scale.

8. Replay modes

Mode	Entry	Use
Broker-like	live/broker_like_replays.py	Cross-strategy leaderboard
Delayed-arming strict	internal replay driver	Promotion standard for delayed-entry gates
Random gate null	internal null-replay driver	Empirical null - randomize prerequisite events only
Hardened realism	live/replay_realism.py	hardened_replay_engine_kwargs()
Signal-only	live/signal_replays.py	Ranking without full broker (weaker)

Primary null (2026-06-26): 200-seed stratified_fine_buckets - year, side, time bucket, range-width quartile matched. Results are kept in internal replay output folders.

9. Metric calculation

Unit / campaign audit

Module: live/replay_audit.py

- Campaign = grouped fills -> units with entry/exit prices.
- Net USD, closed drawdown, intrabar stress DD (adverse path through open units).
- Net/Stress = net / |stress DD|.

Institutional metrics

Script: scripts/institutional_strategy_metrics.py

Daily equity curve -> Sharpe, Sortino, Calmar, QQQ correlation. Output: live/state/institutional_strategy_metrics/metrics.csv.

Inference convention (target)

Series	Use	Status
Campaign P&L (unit_trades, by trade_id)	Primary PSR/DSR, HAC, MinBTL	Phase 1a
Daily equity returns	Secondary exhibit	Current scorecard default

Point values per instrument in replay_audit.POINT_VALUES.

10. Report generation

Scorecard: scripts/generate_strategy_validation_scorecard.py -> live/state/strategy_validation_scorecard/

Outputs: SCORECARD_REPORT.md, index.html, scorecard_data.json, one-page validation pitch drafts, and IMPLEMENTATION_STATUS.md.

Replay artifacts: Major drivers write SUMMARY.md, summary.csv, run_manifest.json, run_manifest.sha256, and per-strategy states/<id>/ folders. Hardened/Tier-1 states include fills, orders, equity/audit outputs, feature_snapshots.csv, and causality_violations.csv.

11. Validation hooks

Hook	Path
DSR trial ledger	data/validation/dsr_trial_ledger.csv
DSR spec	data/docs/DSR_PEER_TECHNICAL_SPEC.md
Causal graph	live/specs/CAUSAL_GRAPH.md
Audit tracker	data/docs/AUDIT_TRACKER.md
Execution scrutiny	internal execution-scrutiny output folder

12. Known concerns register

Concern	Mitigation	Code / artifact	Residual risk	Audit ID
OHLC fill non-uniqueness	Stop-first, gap-through, realism tests	broker.py, broker_realism_validation.py	Model-candle proof missing	1.1
Lookahead / causality	live_after ts, gate events, FeatureSnapshot, CausalityGuard	broker.py, manager.py, causality.py, Tier-1 plugins	Feature snapshots are opt-in, older/non-Tier-1 plugins may only have order-time checks	CO1/CO2
Same-minute ambiguity	Scrutiny + tick manifest	execution-scrutiny driver	Tick recon incomplete	1.4
Multiple testing	DSR N_eff, gate null	scorecard, random gate replay	Scorecard not yet wired to null	FD3
Daily vs campaign inference	Campaign PSR primary (Phase 1a)	scorecard	Daily SR still in metrics CSV	P2
Stratified bucket definition	stratified_fine_buckets documented	AUDIT_TRACKER FD5	Spec coarse buckets not yet run	FD5
Peer benchmark	Suppress if NA	peer_comparison_table.csv	No sourced peer data	P5
Market-specific validation gap	Exclude until source restored	internal output index	Missing null row for one market	Parity table

13. Extension guide

New StrategyPlugin

1. Add class under live/strategies/, register in registry.py.
2. Emit intents with live_after_ts = confirming bar timestamp.
3. Add broker-like row to broker_like_replays.py or dedicated driver.
4. Log DSR ledger row before reviewing results.
5. Update this doc §5 table and CAUSAL_GRAPH.md falsification row if Tier-1.

New null family

1. Copy the internal null-replay pattern - randomize only the hypothesized edge input; keep engine + broker fixed.
2. Freeze strata metadata JSON before seeds run.
3. Set counts_toward_permutation_test in summary CSV.
4. Wire into scorecard empirical p-value block.

Maintenance

Any change to fill semantics, causality guards, or reported metrics must update this file in the same PR as code/tests.